National University
of Computer & Emerging Sciences

EE223 Signals and Systems
Midterm-2

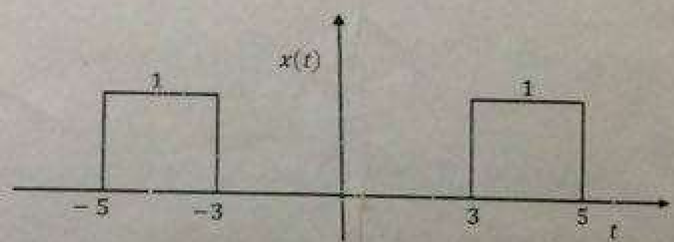
Time: 60 minutes, Max. Marks: 40, Weightage: 20%, Spring-2015, 05-May

Instructions:

Attempt all the questions.
All questions carry equal marks.

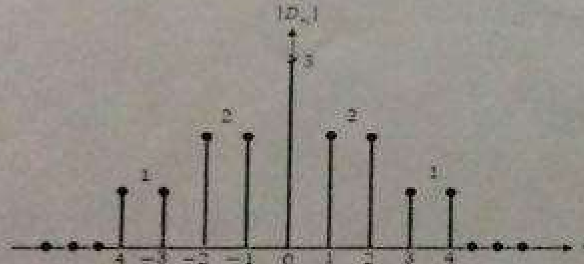
Question # 1: **10 Marks**

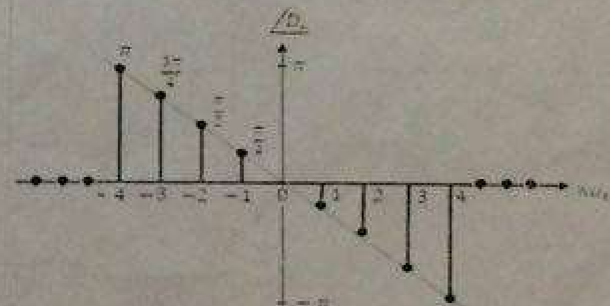
Find the Fourier Transform of following signal .



Question # 2: **10 Marks**

Exponential Fourier Spectra of a periodic signal $x(t)$ are given below:





a) Find the exponential Fourier series using the above spectra.

b) Find the compact trigonometric Fourier series of $x(t)$.

c) Sketch the spectra of compact trigonometric Fourier series using part (b).

Question # 3: 10 Ma

Fourier Transform of a continuous time signal $x(t)$ is given below:

$$X(\omega) = 1 - e^{-j\omega}$$

Using properties, find the mathematical expression for the Fourier Transform of following signals.

Note: Clearly explain which property you are using in each part.

a) $x(t-6)$
b) $x(t)\cos 2\pi t$
c) $x\left(\frac{t}{4}\right)$
d) $\frac{dx(t)}{dt}$

Question # 4: **10 Marks**

A continuous time signal $y(t)$ is defined as $y(t) = x_1(t) * x_2(t)$ where

$x_1(t) = sinc 4\pi t$ and $x_2(t) = \cos 2\pi t$

a) Sketch $x_1(t)$ and $X_1(\omega)$
b) Sketch $x_2(t)$ and $X_2(\omega)$
c) Sketch $Y(\omega)$

Note: Please label each graph completely.